Multisensor Fusion for 3D Facial Landmark Detection in Unilateral Cleft Lip

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***Abstract*— Three-dimensional facial imaging fuses photometric and geometric sensor streams to enable quantitative assessment of facial soft-tissue morphology, offering significant potential for surgical planning and outcome evaluation in patients with unilateral cleft lip (UCL). However, these applications depend on accurate and reproducible landmark identification, which remains limited by the time demands and observer variability of manual 3D scan annotation. We present a point-transformer-based method that fuses RGB texture, surface geometry, multi-scale curvature, and bilateral symmetry descriptors into a unified per-vertex representation, and predicts clinically relevant landmarks directly on external 3D facial meshes from 20 patients with UCL, avoiding 2D projection and preserving the full 3D surface geometry and texture. Landmarks were selected based on Fisher's anatomical subunit approximation technique for unilateral cleft lip repair. Preliminary validation on a subject-wise held-out split achieved a mean landmark error of 2.17 mm and a PCK@5 mm of 96.9%, suggesting promising localization of nasal, nasolabial, and lip landmarks across subjects with variable cleft-related anatomy.**

***Index Terms*— 3D facial mesh, multisensor fusion, RGB-D facial imaging, point transformer, landmark detection, cleft lip**

Code Availability

Code is available at: `https://github.com/Layal-Bou-Hamdan/cleftlip`

I. Introduction

The accurate location of anatomical and surgical landmarks is essential for image-based quantitative evaluation, treatment planning, and postoperative evaluation in patients with cleft lip. These landmarks underlie measurements of facial form, symmetry, growth, and treatment outcomes, making them central to clinical decision-making and craniofacial research [1], [2]. However, capturing them reliably depends on imaging modalities capable of faithfully representing the complex three-dimensional geometry of the face.

Three-dimensional facial imaging meets this need by recording surface morphology without ionizing radiation and providing a more complete representation of facial shape than two-dimensional photographs [3]. Modern 3D stereophotogrammetry systems simultaneously capture two distinct sensor streams: a photometric stream yielding high-resolution RGB texture, and a geometric stream yielding dense surface depth and mesh topology. Effectively fusing these complementary modalities is essential, because texture alone cannot resolve ambiguous surface geometry near the cleft margin, and geometry alone discards appearance cues that distinguish anatomically similar regions. Consequently, it has become the modality of choice in cleft lip research, allowing the evaluation of facial and palatal shape, treatment outcomes, and developmental differences among patient groups [2], [4].

Landmark annotation on 3D meshes is still most often performed manually, a process that is time-consuming, observer-dependent, and particularly vulnerable to the irregular anatomy characteristic of cleft patients, where scans may contain holes, missing regions, surface noise, or inconsistent head pose [5].

Moreover, subtle differences in facial symmetry, lip morphology, and nasal shape can significantly affect functional and aesthetic results, yet the traditional assessment is commonly based on subjective clinical judgment [6]. Together, these factors motivate automated 3D landmark detection methods that reduce annotation time, limit observer dependence, and support more reproducible quantitative analysis of cleft-related facial morphology [6], [7].

This study develops a direct 3D learning approach for predicting clinically significant external facial landmarks on patient-specific facial surface meshes. The method treats landmark detection as a multisensor fusion problem: per-vertex representations are constructed by integrating photometric appearance, surface geometry, intrinsic curvature descriptors, and clinically motivated symmetry scores before being passed to a point-transformer network. The broader goal is to support scalable and objective craniofacial assessment.

II. Related Work

Deep learning has transformed landmark prediction by learning discriminative features directly from data, rather than relying solely on handcrafted descriptors or rule-based pipelines. Within biomedical image analysis, convolutional architectures such as U-Net have become standard tools for localization and segmentation tasks [8]. Similarly, in facial analysis, deep neural networks have improved the robustness of face alignment and landmark prediction across diverse images and subjects [9]. These advances are particularly

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relevant to 3D craniofacial analysis, where accurate landmark localization is essential for quantifying facial morphology, asymmetry, growth, and treatment outcomes.

Before the widespread adoption of deep learning, early approaches to 3D surface analysis depended heavily on registration-based methods. In this context, Iterative Closest Point and its variants became standard tools for aligning surfaces prior to anatomical comparison [10], [11]. Building from this registration-based foundation, later work introduced more specialized architectures for 3D facial landmark detection, including CNN-based surface representations, heatmap regression, object-detection strategies, graph convolutional networks, and direct soft-tissue landmark prediction [12]–[15].

Alongside these developments, PointNet and PointNet++ further advanced 3D analysis by enabling direct processing of unordered point sets, thereby removing the need to convert 3D geometry into images or voxels [16], [17]. More recently, transformer-based point-cloud models have extended this direction by using attention mechanisms to model relationships across local and global geometric structures [18].

Artificial intelligence has also begun to play a broader role in cleft care, with reported applications in diagnosis, treatment planning, presurgical orthopedics, speech evaluation, surgery, and outcome prediction [19], [20]. Within external facial landmark detection, Sayadi et al. demonstrated AI-based detection of surgically important landmarks on nasolabial images of patients with unilateral cleft lip [21], while Byun et al. proposed automated landmark detection for an augmented-reality craniofacial surgical assistance system [22].

Despite these advances, many cleft-specific automated landmark prediction studies have remained focused on intraoral, presurgical, or CT-derived structures rather than on external facial soft-tissue geometry. For example, Schnabel et al. developed an automated pipeline for presurgical plate treatment [5], Agaronyan et al. proposed a geometric deep learning approach for maxillary-arch landmark detection on 3D oral scans of newborns with cleft lip [23], and Xu et al. addressed 3D cephalometric landmarking from CT-derived point clouds [24]. Taken together, these studies demonstrate the growing value of automated analysis in cleft populations. However, relatively few studies have examined direct external facial soft-tissue landmark prediction on 3D photogrammetric meshes of patients with UCL. The present study addresses this gap by predicting anatomical landmarks directly from external 3D facial surface meshes using a multisensor fusion of photometric and geometric modalities, without converting the data into 2D rendered views or relying on intraoral representations.

III. Method

A. Data Collection and Landmark Annotation

The dataset consisted of 3D photogrammetry facial scans from 20 patients with unilateral cleft lip (UCL). The scans were collected to study external facial differences associated with UCL and captured the visible soft-tissue surface of the face rather than intraoral palatal anatomy. All data were de-identified prior to transfer and used under institutional approval and an approved Data Use Agreement. The data were obtained through collaboration between the University of Tennessee Health Science Center (UTHSC) and the University of Tennessee Knoxville (UTK), and were transferred for model development under an approved Data Use Agreement. Each scan was exported as a textured 3D facial surface mesh in OBJ format.

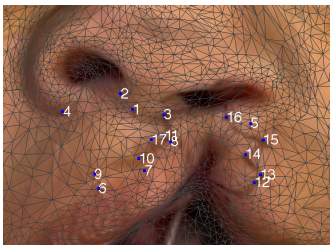


Fig. 1. Annotated 17 anatomical landmarks on the 3D facial mesh.

A plastic and reconstructive surgeon provided clinical guidance for landmark selection. A total of 23 external facial landmarks were manually annotated on each mesh. Of these, 17 are surgically relevant landmarks selected based on Fisher's anatomical subunit approximation technique for unilateral cleft lip repair [25], focusing on clinically important nasal, nasolabial, upper-lip, and perioral soft-tissue regions; these 17 landmarks serve as the direct prediction targets for model training and evaluation. The remaining 6 landmarks comprise the 4 bilateral inner and outer eye canthi and the 2 mouth corners, which are not surgical targets but provide stable anatomical reference points used for mesh alignment, intercanthal scaling, and symmetry computation. The complete annotated set of 17 landmarks is illustrated in Fig. 1.

B. Preprocessing

Raw clinical meshes vary in orientation, scale, density, and surface quality, and many contain noise, holes, and irregular geometry around the cleft region. The preprocessing pipeline standardizes each scan while preserving clinically relevant cleft-related nasolabial morphology. It includes robust template-based alignment using a coarse-to-fine ICP framework with FPFH-RANSAC initialization [10], [11], [26], boundary removal through adaptive spherical cropping centered at the mesh centroid, and data augmentation using left-right reflection, rigid pose perturbation, semantically constrained non-rigid deformation of nasal and lip regions, and photometric texture variation.

Landmark coordinates are transformed using the same operations applied to the mesh vertices, ensuring that spatial registration is maintained throughout preprocessing. Quality filtering then retains only alignments with bidirectional symmetric surface overlap exceeding 88%. Full details of all preprocessing parameters are provided in Appendix I.

C. Multisensor Feature Fusion

A core contribution of this work is the construction of a per-vertex feature representation that fuses four complementary sensor-derived and task-specific information streams into a unified 19-dimensional encoding. Integrating these streams is motivated by the limitations of any single modality: geometric descriptors resolve surface shape but are blind to appearance-based tissue boundaries, while photometric features encode color and texture but cannot represent 3D curvature. The fused streams are expected to provide complementary cues: RGB texture helps identify soft-tissue appearance variation, geometric and curvature descriptors capture local and multi-scale surface shape, and the bilateral symmetry score provides a task-specific cue for cleft-related asymmetry.

(i) Geometric stream (position and normals). The 3D Cartesian position $\mathbf{x} \in \mathbb{R}^3$ and unit vertex normal $\mathbf{n} \in \mathbb{R}^3$ are derived directly from the depth and surface sensor output, encoding spatial location and local surface orientation.

(ii) Photometric stream (RGB texture). The color $\mathbf{c} \in \mathbb{R}^3$ is sampled from the high-resolution texture map captured by the stereophotogrammetry camera array, encoding appearance cues such as skin tone variation, lip vermilion boundaries, and scar tissue.

(iii) Intrinsic surface descriptors. Multi-scale principal curvatures $k^r_{\min}$ and $k^r_{\max}$ at neighborhood scales $r \in \{10, 30, 100\}$, a Heat Kernel Signature descriptor h [27], the distance to the mesh centroid d_c, and mean local geodesic-neighbor distance g collectively characterize local and global surface geometry at multiple resolutions, capturing anatomical shape detail that neither positional nor photometric features alone can resolve.

(iv) Bilateral symmetry score. The symmetry term s explicitly encodes the left-right asymmetry that is the defining clinical characteristic of UCL morphology. After alignment to the intercanthal reference frame, each vertex $\mathbf{x}_i$ is reflected across the estimated facial midline plane to obtain $\mathbf{x}_i^{\text{ref}}$. The score is computed as the distance from this reflected point to the closest observed surface point on the opposite side,

$$s_i = \min_{\mathbf{x}_j \in \mathcal{V}_{\text{opp}}} \|\mathbf{x}_i^{\text{ref}} - \mathbf{x}_j\|_2,$$

optionally normalized by the intercanthal distance. Larger values indicate stronger local deviation from bilateral symmetry.

The complete fused feature vector is ordered as

$$[\mathbf{x},\ \mathbf{c},\ \mathbf{n},\ d_c,\ k^{10}_{\min},\ k^{10}_{\max},\ k^{30}_{\min},\ k^{30}_{\max},\ k^{100}_{\min},\ k^{100}_{\max},\ h,\ s,\ g].$$

D. Graph Construction

Mesh connectivity is encoded as a directed graph with multi-hop neighborhoods. Importantly, the base neighbor graph is not built from Euclidean k-nearest neighbors. Instead, 1-hop neighbors are derived from mesh topology where two vertices are adjacent if they share a triangular face. This choice avoids artificial shortcuts across nearby-but-disconnected facial regions, such as opposing lip surfaces or narrow gaps created by scan artifacts. The face-adjacency list is padded to a fixed width for storage, and directed edge sets are then expanded to k-hop connectivity with $k \in \{1, 2, 3\}$. The resulting k-hop graphs include all vertices reachable within k topological steps, with self-loops removed and duplicate edges collapsed. This graph construction better respects the underlying facial surface than Euclidean neighbor expansion.

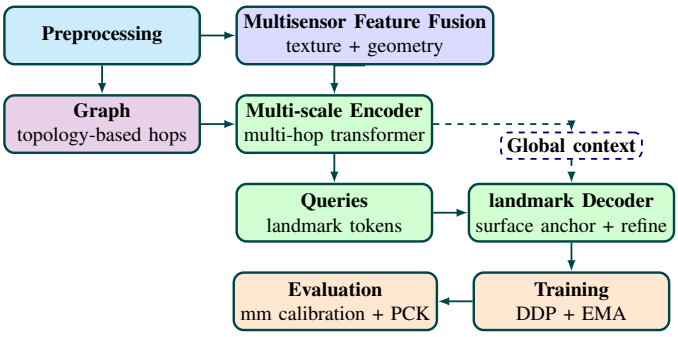


Fig. 2. Overview of the cleft lip landmark pipeline. Preprocessing covers registration, mesh cleanup, and augmentation. Each mesh is then converted into per-vertex appearance and geometric descriptors and a topology-aware graph. A multi-hop encoder produces mesh features and global context, landmark-query tokens drive a surface-anchored decoder, and the resulting model is optimized with distributed training and evaluated in millimeters using calibrated PCK-based metrics.

E. Model Architecture

The overall network architecture is depicted in Fig. 2. It consists of three main functional stages: input projection and positional encoding, multi-scale point convolution, and landmark-specific decoding.

Of the 23 landmarks annotated on each mesh, 17 surgically relevant landmarks are used as direct prediction targets and contribute to the training loss. The remaining 6 landmarks, comprising the 4 bilateral inner and outer eye canthi and the 2 mouth corners, are propagated to all subject meshes via the ICP-computed transformation but are excluded from loss computation. They serve as stable reference points for alignment verification, intercanthal scaling, and symmetry score computation.

1) Point Transformer Network Design: **Input projection.** The raw 19-dimensional fused vertex features are concatenated with Fourier positional encodings of the 3D coordinates across six frequency bands and projected through a two-layer MLP with GELU activations, layer normalization, and dropout into a common latent space.

Multi-scale encoder. Six stacked point-transformer blocks apply geometry-aware graph convolution with a scheduled neighborhood sequence $[1, 1, 2, 2, 3, 3]$, so early layers capture local surface detail while later layers integrate broader facial context. This multi-scale aggregation mirrors a spatial sensor fusion strategy. Fine-scale neighborhoods fuse local curvature and texture cues, while coarse-scale neighborhoods fuse global facial shape and symmetry context. Each block follows a residual transformer-style structure with layer normalization, point-transformer convolution, a feed-forward

MLP, and stochastic depth regularization with linearly increasing DropPath rate.

Landmark decoder. 17 learnable landmark tokens attend over encoded vertices to produce a surface-anchored initial proposal:

$$\mathbf{p}_0 = W\mathbf{P},$$

where $\mathbf{P}$ is the matrix of vertex coordinates and W contains the token-to-vertex attention weights. A 2-layer transformer encoder with multi-head self-attention then refines the tokens, allowing each landmark to interact with global facial structure and with the positions of other landmarks. Final predictions are formed as

$$\hat{\mathbf{p}} = \mathbf{p}_0 + \Delta\mathbf{p},$$

where $\Delta\mathbf{p}$ is a residual offset produced by a lightweight MLP head. This surface-anchored formulation reduces the regression burden. The model begins from a soft point on the observed surface and learns only the local correction instead of inferring absolute coordinates from a global latent code.

2) Training and Evaluation: Before augmentation, the 20 subjects were divided at the patient level into training, validation, and test sets using a 70/20/10 split, corresponding to 14 training subjects, 4 validation subjects, and 2 test subjects. All mirrored and augmented variants generated from a given subject were kept within the same partition as the original subject. This subject-wise split prevents leakage of patient-specific anatomy across training, validation, and test sets.

The model is trained end-to-end using Smooth L1 (Huber) loss on landmark coordinates, defined as

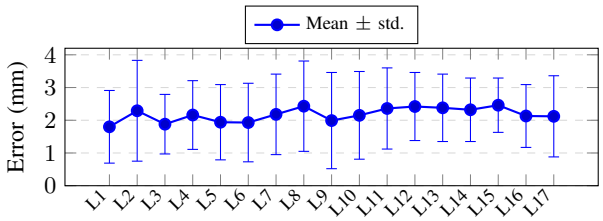
$$\mathcal{L}_{\text{Huber}}(e) = \begin{cases} \frac{1}{2}e^2 & \text{if } |e| \le \beta \\ \beta\left(|e| - \frac{1}{2}\beta\right) & \text{otherwise,} \end{cases}$$

where e is the error between predicted and ground-truth landmark coordinates and $\beta = 2.0$, with an orthogonality regularizer weighted at 10^{-3} on the learnable landmark tokens to maintain distinct landmark identities. Full training configuration is provided in Table II in the appendix.

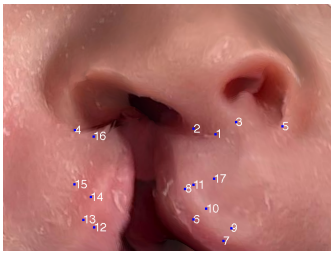
Evaluation uses per-landmark Euclidean error in millimeters and Percentage of Correct Keypoints (PCK) at 2 mm, 3 mm, 5 mm, and 10 mm thresholds. All experiments were conducted on 4 NVIDIA H100 GPUs with 80 GB memory each.

IV. Results

The results in Fig. 3a show per-landmark mean errors ranging from 1.80 mm to 2.46 mm, indicating comparatively uniform localization accuracy across the 17 predicted landmarks. The observed standard deviations suggest that some landmarks are more variable across subjects, likely because many points lie near the nasal base, lip surface, and cleft margin. These regions have complex geometry and subject-specific anatomical variation, making precise landmark localization more challenging. Compared with related methods, this performance is consistent with the 2–3 mm range reported for surface-based craniofacial landmark detection



(a) Mean per-landmark error with standard deviation.



(b) Landmark locations used for evaluation.

Fig. 3. Per-landmark localization performance and corresponding landmark locations. Errors are reported in millimeters.

in challenging small-dataset settings [5], [23]. Clinically, an average error near 2 mm is promising for preliminary quantitative assessment of nasal, nasolabial, and lip morphology, although its suitability for surgical decision-making should be confirmed through larger studies and surgeon-in-the-loop validation. The result was obtained using the full fused feature representation, which combines RGB texture, surface geometry, multi-scale curvature descriptors, and bilateral symmetry information directly on the native mesh surface without 2D projection. Preliminary feature-drop experiments further supported this design choice: removing individual descriptor groups from the fused input increased validation error relative to the full model. Because of the small validation set, these ablation results are interpreted qualitatively rather than as definitive estimates of each stream's independent contribution and will require confirmation on a larger independent dataset.

V. Conclusion

We presented a point-transformer-based method for automated detection of clinically relevant external facial landmarks on 3D photogrammetric meshes from patients with unilateral cleft lip. The proposed pipeline frames landmark detection as a multisensor fusion problem, combining RGB texture, surface geometry, multi-scale curvature descriptors, and bilateral symmetry scores into a unified per-vertex representation before feeding them to a point-transformer network. This approach avoids 2D projection and operates directly on the facial surface geometry, preserving spatial structure throughout preprocessing, feature extraction, and prediction. The surface-anchored decoder, which regresses residual offsets from attention-weighted vertex proposals, reduces the regression burden and supports stable learning on the small UCL dataset.

The model achieved a mean landmark error of 2.17 mm,

with per-landmark errors ranging from 1.80 mm to 2.46 mm and an average PCK@5 mm of 96.9%. Although these preliminary results are limited by the small dataset size, they suggest that direct 3D mesh-based learning using a fused multimodal per-vertex representation can support localization of surgically important nasal, nasolabial, and lip landmarks in subjects with irregular cleft-related anatomy.

Several limitations and directions for future work remain. The dataset is small, and expanding it to include more patients, age groups, and postoperative scans would improve generalization and support longitudinal outcome assessment. In future work, we will use the predicted landmarks to perform geometric analysis of patient outcomes, including facial symmetry scoring to estimate treatment effectiveness. We will also develop a visualization platform to display landmark placement on 3D facial images, improve model interpretability, and support clinician review of the predicted anatomical regions. Additional extensions may include expanding the landmark set to intraoral and skeletal structures using CT or CBCT data, integrating complementary sensor modalities such as structured light or CBCT, and conducting prospective surgeon-in-the-loop validation.

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APPENDIX I
PREPROCESSING DETAILS

A. Robust Template-Based Alignment

Each subject mesh was aligned to a high-quality reference template using an improved ICP framework [10]. Alignment began with landmark-based prescaling to match the template inner intercanthal distance. Global registration was then initialized using FPFH-RANSAC [26], with a small multi-scale search around the nominal voxel size retaining the solution with highest feature-matching fitness. Multiple rotation hypotheses were evaluated alongside the identity pose to prevent anatomically implausible convergence, including flipped or inverted alignments.

The best initialization was refined using a four-stage coarse-to-fine ICP strategy [11]. Early stages used stronger downsampling and point-to-point matching for stable coarse registration; later stages used finer sampling and normal estimation for local surface alignment. Anatomical validation used the left-to-right eye-vector cosine similarity between transformed subject and template landmarks to detect failed alignments.

B. Boundary Removal

Following alignment, noisy boundary regions such as ears, neck edges, and scan artifacts are removed by cropping to a sphere centered at the mesh centroid. The crop radius is set adaptively at the 80th percentile of all vertex distances from the centroid, removing the 20% farthest vertices while preserving core facial geometry.

C. Data Augmentation

To address limited dataset size and balance laterality, each preprocessed mesh is first mirrored across the intercanthal plane, converting left-sided cleft morphology to right-sided and vice versa, with face winding reversed to preserve outward-pointing normals. This doubles the number of base meshes.

Additional variants are generated using rigid pose perturbation, local geometric deformation, optional mesh jitter, and photometric texture variation. Mouth deformation pulls vertices near mouth landmarks toward a midpoint between upper and lower lip regions. Nose deformation applies random directional translation to vertices near nasal landmarks, simulating cleft-related nasal asymmetry. Optional Gaussian vertex jitter breaks artificial geometric regularity after parametric deformation. RGB texture maps undergo probabilistic brightness and contrast perturbation. Landmarks can be projected onto the nearest mesh surface after deformation to ensure geometric consistency.

D. Preprocessing Settings

TABLE I
PREPROCESSING SETTINGS USED FOR MESH AUGMENTATION.

Parameter	Value
Strength range	0.01–0.45
Radius range	0.05–0.40 of bbox diagonal
Rotation range	$\pm 40^{\circ}$
Translation range	± 0.05
Grid size	12 strengths $\times$ 12 radii
Mesh jitter std.	0.0001

TABLE II
TRAINING CONFIGURATION FOR THE POINT-TRANSFORMER LANDMARK DETECTOR.

Component	Setting
Fourier positional encoding	6 bands, max frequency 2.0
Embedding dimension	128
Decoder refinement	2-layer transformer encoder, 8 heads
Loss	Smooth L1 (Huber), $\beta = 2.0$
Orthogonality weight	10^{-3}
Optimizer	AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.98$
Initial learning rate	10^{-4} (scaled with global batch size)
Weight decay	10^{-4}
Epochs	300
Warm-up	5 epochs linear
LR scheduling	ReduceLROnPlateau
Dropout	Attention 0.25, MLP 0.35
DropPath	0.2 max
LayerScale init	10^{-4}
EMA decay	0.999
Batch size / GPUs	4 per GPU on 4 GPUs

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